Experiment No 4

Course Outcome: CO2 **Blooms Level: L3**

Aim: To explore data visualization techniques on the given dataset.

Abstract: The problem is to perform exploratory data analysis on the Seaborn Tips dataset by applying suitable data visualization techniques in order to understand the underlying structure and behavior of the data.

- The dataset consists of numerical attributes such as total bill and tip amount, along with categorical attributes including day of the week, time of dining, gender, and smoking status.
- The task is to identify and apply appropriate visualization methods to represent frequency distributions of categorical variables and distribution characteristics of numerical variables.
- Visualizations are to be used to detect patterns, variations, and possible outliers present in the dataset.
- Proper selection of plots is required to ensure clear and effective representation of different data types.

The problem further aims to analyze relationships among multiple variables using comparative and relational visualization techniques.

- The relationship between total bill and tip amount is to be examined to understand tipping trends.
- The effect of categorical factors such as day, time, gender, and smoking status on tipping behavior is to be analyzed through comparative plots.
- Multi-variable visualizations are to be used to observe combined effects and interactions among attributes.
- The final objective is to interpret the visual outputs and draw meaningful inferences that support data-driven understanding and exploratory data analysis skills.

Theory: Visualizations make it easier to explore and extract relevant information from the data by identifying patterns, relationships, outliers, and much more. Seaborn is a statistical plotting library in Python and is an extended version of Matplotlib. It supports complex visualizations and makes the plotting of graphs simple and intuitive. It can be used in Python scripts, Jupyter notebook, and web application servers.Seaborn uses less syntax as compared to Matplotlib. Hence, it is easier to use.It is easier to customize themes and high-level interfaces in Seaborn to make the plots more attractive and readable.Seaborn is much more functional and organized than Matplotlib and is better integrated to work with Pandas DataFrames.

Seaborn provides different plots for different types of variables as follows:

a. Frequency Distribution - Categorical Variables
* countplot
* catplot

b. Distribution of the Numerical Variable
* distplot(histogram)
* kdeplot
* boxplot

* violinplot

c. Relationship between 2 Numerical Variables
* lineplot
* scatterplot
* relplot
* lmplot
* heatmap
* pairplot
* facetgrid

d. Relationship between Numerical and Categorical Variables
* pointplot
* barplot
* boxplot
* violinplot
* swarmplot
* catplot
* facetgrid

Program:

Importing required libraries

In [1]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from scipy import stats
%matplotlib inline
import warnings
warnings.simplefilter(action='ignore', category=FutureWarning)
```

```
C:\Users\Ramya\Anaconda3\lib\site-packages\statsmodels\tools\_testing.py:19: FutureWarning: pandas.ut
il.testing is deprecated. Use the functions in the public API at pandas.testing instead.
  import pandas.util.testing as tm
```

Loading dataset

In [2]:
```
tips=sns.load_dataset('tips')
df=pd.DataFrame(tips)
df.head()
```

Out[2]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Preprocessing and Exploratory Data Analysis

checking for missing data

In [3]:
```
df.isnull().sum()
```

Out[3]:
```
total_bill    0
tip           0
sex           0
smoker        0
day           0
time          0
size          0
dtype: int64
```

Viewing the descriptive statistics of the dataset

In [4]:
```
df.describe()
```

Out[4]:

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

Get a numerical summary for 'tip'

In [5]:
```
df.tip.describe()
```

Out[5]:
```
count    244.000000
mean       2.998279
std        1.383638
min        1.000000
25%        2.000000
50%        2.900000
75%        3.562500
max       10.000000
Name: tip, dtype: float64
```

Five Number Summary For bill and tip

In [6]:
```
bill = df.total_bill

print("Maximum Bill = ",np.max(bill))
print("Minimum Bill = ",np.min(bill))
print("Standard Deviation = ",np.std(bill))
print("Median =",np.median(bill))
print("Mean = ",np.mean(bill))
```

```
Maximum Bill =  50.81
Minimum Bill =  3.07
Standard Deviation =  8.88415057777113
Median = 17.795
Mean =  19.785942622950824
```

In [7]:
```
tip = df.tip
print("Maximum Tip = ",np.max(tip))
print("Minimum Tip = ",np.min(tip))
print("Standard Deviation = ",np.std(tip))
print("Median =",np.median(tip))
print("Mean = ",np.mean(tip))
```

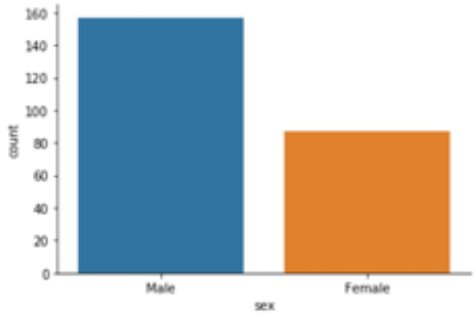
```
Maximum Tip =  10.0
Minimum Tip =  1.0
Standard Deviation =  1.3807999538298958
Median = 2.9
Mean =  2.9982786885245902
```

Exploratory Data Analysis

Explore if there is any dependency between the variable "Tip" and rest of the variables.

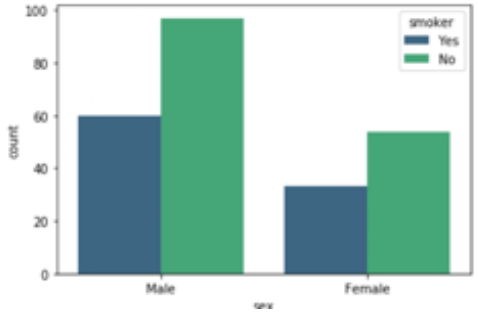
In [29]:
```
sns.countplot(x='sex', data=tips)
sns.despine() # no top and right axes spine

print(tips.sex.value_counts())
```

```
Male      157
Female     87
Name: sex, dtype: int64
```



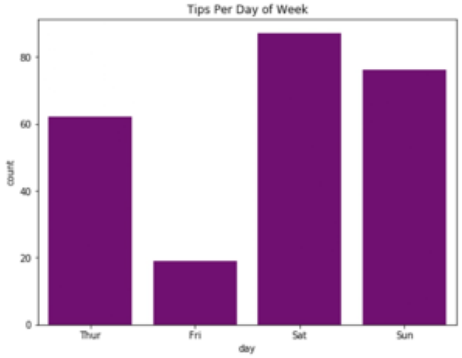
In [9]:
```
sns.countplot(x='sex', data=tips, hue='smoker', palette='viridis')
```

Out[9]:
```
<matplotlib.axes._subplots.AxesSubplot at 0x1e6ee9b5588>
```



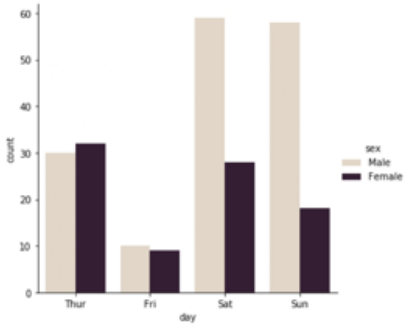
In [10]:
```
plt.figure(figsize=(8,6))
plt.title("Tips Per Day of Week")
sns.countplot(x=tips['day'],color = 'purple')
```

Out[10]: <matplotlib.axes._subplots.AxesSubplot at 0x1e6eea6f108>



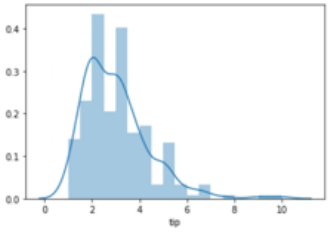
In [11]:
```
sns.catplot(x='day', data=tips, hue='sex', palette='ch:.25', kind='count')
```

Out[11]: <seaborn.axisgrid.FacetGrid at 0x1e6eea6bd88>

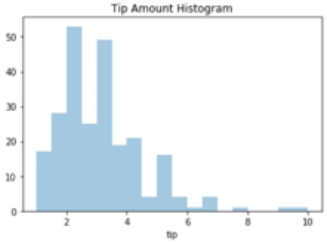


In [12]:
```
sns.distplot(df['tip'])
```

Out[12]: <matplotlib.axes._subplots.AxesSubplot at 0x1e6eeada4c8>



In [13]:
```
g = sns.distplot(tips.tip,kde=False)
g.set_title('Tip Amount Histogram');
```

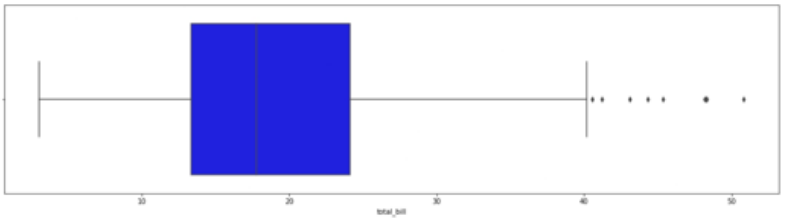


Find out the outliers for "Total Bill" and "Tip"

outliers in bill column

In [14]:
```
plt.figure(figsize=(20,5))
sns.boxplot(x=bill, color='b')
```

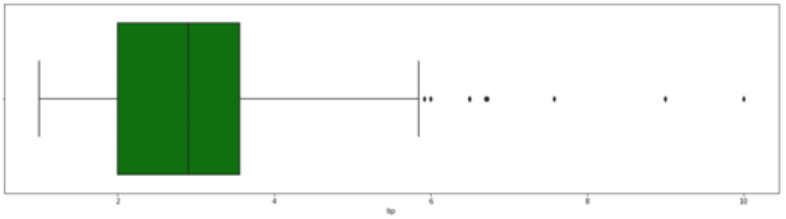
Out[14]:
```
<matplotlib.axes._subplots.AxesSubplot at 0x1e6eec270c8>
```



outliers in tip column

In [15]:
```
plt.figure(figsize=(20,5))
sns.boxplot(x=tip, color='g')
```

Out[15]:
```
<matplotlib.axes._subplots.AxesSubplot at 0x1e6ec937408>
```



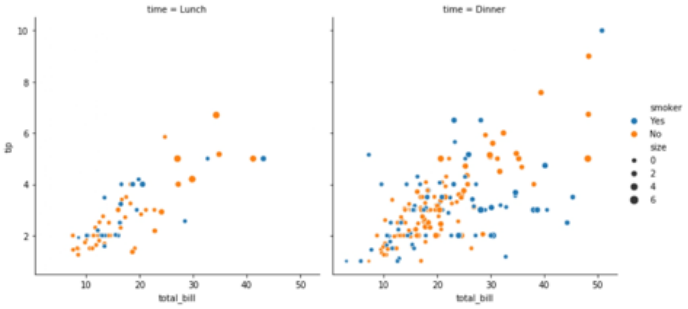
IQR value

In [16]:
```
bill_tip = pd.DataFrame(df,columns=['total_bill','tip','size'])
print(bill_tip)
print("IQR For Total Bill : ",stats.iqr(bill))
print("IQR For Tip : ",stats.iqr(tip))
```

```
     total_bill   tip  size
0         16.99  1.01     2
1         10.34  1.66     3
2         21.01  3.50     3
3         23.68  3.31     2
4         24.59  3.61     4
..          ...   ...   ...
239       29.03  5.92     3
240       27.18  2.00     2
241       22.67  2.00     2
242       17.82  1.75     2
243       18.78  3.00     2

[244 rows x 3 columns]
IQR For Total Bill :  10.779999999999998
IQR For Tip :  1.5625
```

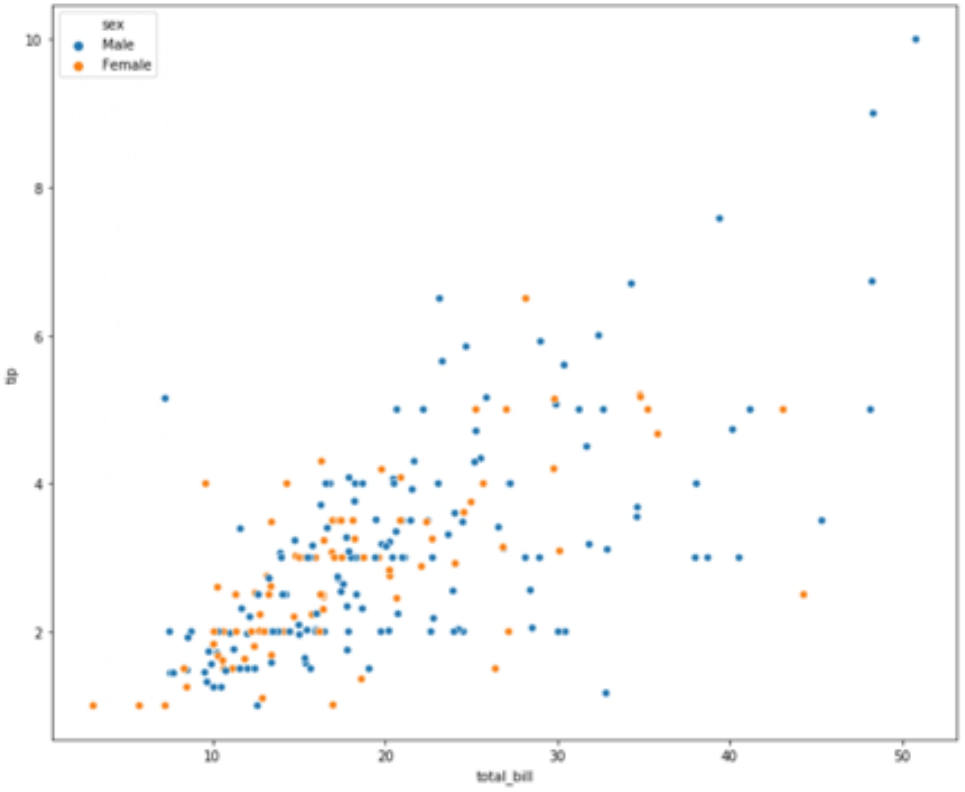
In [17]:
```
sns.relplot(x='total_bill',y='tip',data=df,col='time',hue='smoker',size='size')
```

Out[17]:
```
<seaborn.axisgrid.FacetGrid at 0x1e6eed0cc88>
```



plots based on lunch and dinner based on whether a person is a smoker or not.Can see a linaer pattern ie as total bill increases tip also increases. relplot and lmplot are used for visualizing linear relationships.

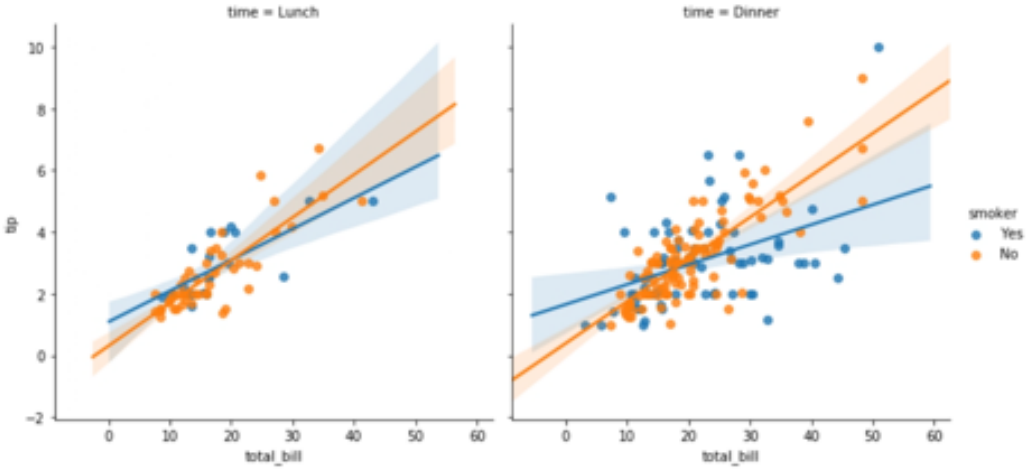
In [18]:
```
plt.figure(figsize=(12,10))
sns.scatterplot(data=df,x="total_bill",y="tip",hue="sex");
```



Can see a linaer pattern ie as total bill increases tip also increases.

In [20]:
```
sns.lmplot(x='total_bill',y='tip',data=df,col='time',hue='smoker')
```

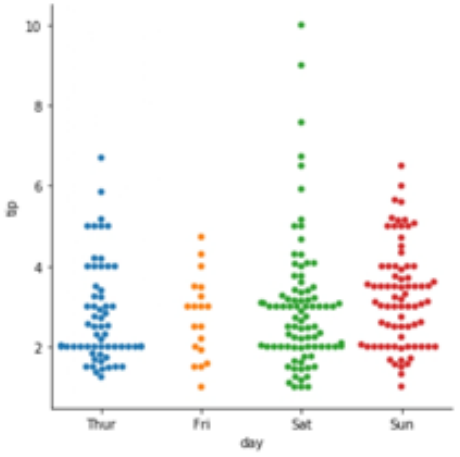
Out[20]: <seaborn.axisgrid.FacetGrid at 0x1e6ef15f888>



If bill amount is less,smokers give more tip than non smokers but they give less tips if the bill amount is high.So it is better to serve a non smoker if the bill amount is high.

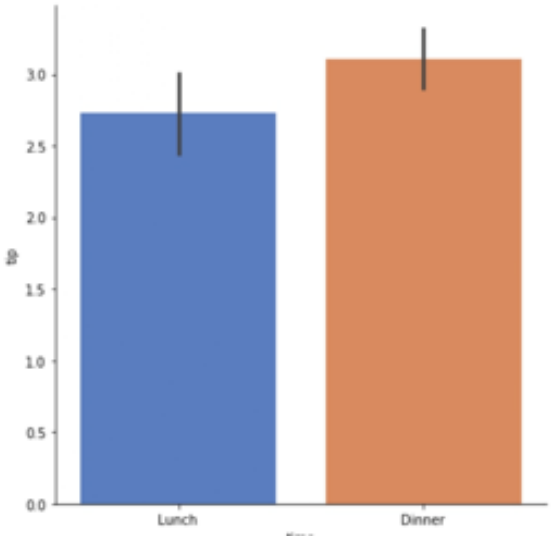
In [21]:
```
sns.catplot(x='day',y='tip',data=df,kind='swarm')
```

Out[21]:
```
<seaborn.axisgrid.FacetGrid at 0x1e6ef24c2c8>
```



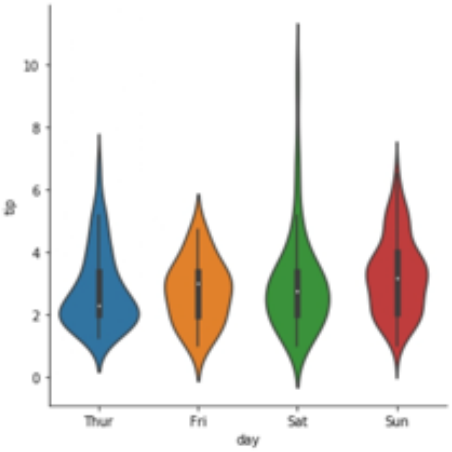
In [22]:
```
sns.catplot(x="time", y="tip", data=df, height=6, kind="bar", palette="muted")
```

Out[22]:
```
<seaborn.axisgrid.FacetGrid at 0x1e6ef45ae08>
```



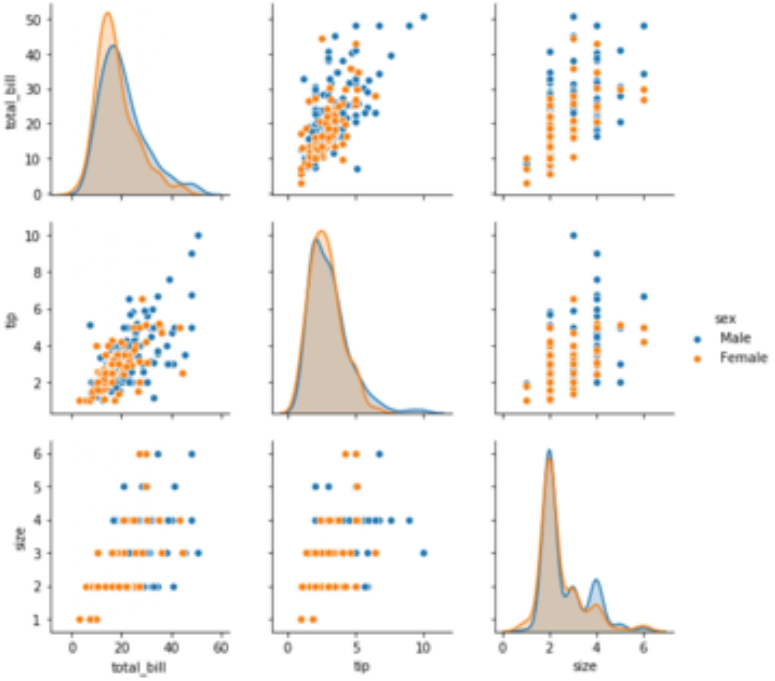
In [23]:
```
sns.catplot(x='day',y='tip',data=df,kind='violin')
```

Out[23]:
```
<seaborn.axisgrid.FacetGrid at 0x1e6ef4a6488>
```



In [24]:
```
sns.pairplot(df, hue='sex')
```

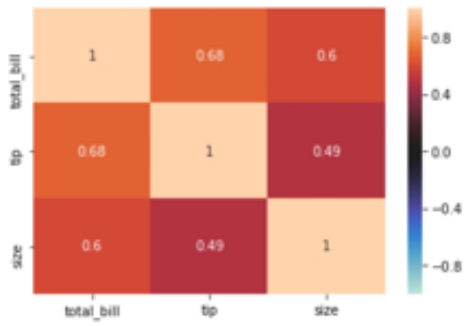
Out[24]:
```
<seaborn.axisgrid.PairGrid at 0x1e6ef53fa08>
```



Correlation Matrix

In [25]:
```
corr_matrix=df.corr()
ax=sns.heatmap(data=corr_matrix,annot=True,vmax=1,vmin=-1,center=0)
bottom, top = ax.get_ylim()
ax.set_ylim(bottom + 0.5, top - 0.5)
```

Out[25]: (3.0, 0.0)



Converting categorical variables into numerical values so that the machine learning model can understand.

In [ ]:
```
from sklearn.preprocessing import LabelEncoder
labelencoder_df=LabelEncoder()
df['sex']=labelencoder_df.fit_transform(df['sex'])
df['smoker']=labelencoder_df.fit_transform(df['smoker'])
df['day']=labelencoder_df.fit_transform(df['day'])
df['time']=labelencoder_df.fit_transform(df['time'])
df.head()
```

Creating heatmap including these numerical values

In [ ]:
```
corr_matrix=df.corr()
ax=sns.heatmap(data=corr_matrix,annot=True,vmax=1,vmin=-1,center=0)
bottom, top = ax.get_ylim()
ax.set_ylim(bottom + 0.5 , top - 0.5)
```

Each square shows the correlation between the variables on each axis. Correlation ranges from -1 to +1. Values closer to zero means there is no linear trend between the two variables. The close to 1 the correlation is the more positively correlated they are; that is as one increases so does the other and the closer to 1 the stronger this relationship is. A correlation closer to -1 is similar, but instead of both increasing one variable will decrease as the other increases. The diagonals are all 1/dark green because those squares are correlating each variable to itself (so it's a perfect correlation). For the rest the larger the number and darker the color the higher the correlation between the two variables. The plot is also symmetrical about the diagonal since the same two variables are being paired together in those squares.

Conclusion: Data visualization is done on the tips dataset of Seaborn using plots for different types of variables and inferences are made about the relationship between total bill, tip, day, time, gender, smoker or non-smoker etc.

Exercise 1: Visualization-Based Analysis of the Titanic Dataset

Abstract:

Using the Seaborn Titanic dataset, perform exploratory data visualization to understand passenger survival patterns.

1. Visualize the frequency distribution of categorical variables such as sex, class, and embarkation port using appropriate plots.
2. Analyse the distribution of numerical variables such as age and fare using suitable distribution plots.
3. Examine the relationship between passenger class and fare using comparative plots.
4. Visualize survival rates with respect to gender and passenger class.
5. Interpret the visualizations and summarize the key insights regarding factors that influenced passenger survival.

Exercise 2: **Visualization-Based Analysis of the Penguins Dataset**

Abstract:

Using the Seaborn Penguins dataset, perform exploratory data visualization to study physical characteristics across different penguin species.

Tasks:

1. Visualize the distribution of numerical attributes such as bill length, bill depth, and body mass.
2. Compare physical characteristics across different species and sex using appropriate categorical–numerical plots.
3. Analyse the relationship between flipper length and body mass using relational plots.
4. Use multivariate visualizations to observe interactions among multiple attributes.
5. Draw meaningful inferences from the visual analysis and discuss observable patterns or differences among species.